

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 8: BOUNDARY DESCRIPTORS

SLO 2: Apply boundary descriptors to extract meaningful features for shape analysis and recognition.

DESCRIPTOR DECISION LAB SOLUTIONS

Case	Boundary Observation	Extracted Feature / Shape Property
1	Boundary has 4 sharp corners	Quadrilateral / Rectangular shape topology
2	Boundary is smooth with no corners	Circular / Elliptical shape continuity
3	Boundary length is very large	High perimeter / Complex or large scale object

Question: Which descriptor property helps identify shape complexity?

Compactness / Perimeter-to-Area ratio & Curvature representation help quantify shape complexity by measuring boundary irregularities and convolutions.

ACTIVITY 2 – ERROR DETECTIVE SOLUTIONS

Detected Feature	System Prediction	Correct Classification
Smooth continuous boundary	Square	Circle (or Ellipse)
Four equal edges	Circle	Square (or Rhombus)
Three corners	Triangle	Triangle (Correctly matched)

1. Identify the incorrect classifications:

The predictions for Smooth continuous boundary → Square and Four equal edges → Circle are swapped and incorrect.

2. Correct the actual shapes based on boundary descriptors:

A smooth continuous boundary corresponds to a Circle, and four equal edges with sharp corners correspond to a Square.

3. Explain which boundary feature was misinterpreted:

The system misattributed boundary continuity (smoothness vs. corner curvature) and linearity (straight edge segments vs. curved contours).